

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think> For part (a), we need to calculate the heat required to raise the temperature of water from 30Â°C to 60Â°C . The formula is $Q = m * c * \Delta T$, where Q is the heat , m is the mass , c is the specific heat , and ΔT is the temperature change . Water's density is approximately 1 g/mL , so the mass of 1 liter of water is 1000 g . The specific heat of water is 1 cal/gÂ°C , and the temperature change is 60Â°C - 30Â°C = 30Â°C . So , $Q = 1000\text{ g} * 1\text{ cal/gÂ°C} * 30\text{Â°C} = 30,000\text{ cal} = 30\text{ Kcal} .$

For part (b), we need to heat a 1 kg block of aluminum from 30Â°C to 60Â°C . The mass is 1000 g , the specific heat of aluminum is 0.215 cal/gÂ°C , and the temperature change is 30Â°C . So , $Q = 1000\text{ g} * 0.215\text{ cal/gÂ°C} * 30\text{Â°C} = 645\text{ cal} = 6.45\text{ Kcal} .$

</think>< answer> 30 Kcal , 6.45 Kcal </answer>

